Т14

- $x_n \sim N(a, \sigma_x^2)$, где $\sigma_x^2 = 2$
- $y_m \sim N(b, \sigma_y^2)$, где $\sigma_y^2 = 1$
- $x = \{-1{,}11;\ -6{,}10;\ 2{,}42\}$ $n=3$

$y = \{-2{,}29;\ -2{,}91\}$ $m=2$

возьмем $\alpha = 0{,}05$

$H_0: a=b$ $\quad H_1: a>b$

1. Точечные оценки матожиданий:

$$\bar{x} = \frac{-1{,}11-6{,}10+2{,}42}{3} \approx -1{,}597$$

$$\bar{y} = \frac{-2{,}29-2{,}91}{2} = -2{,}6$$

2. $H_0: a-b=0$ $\quad \bar{x} \sim N\left(a, \frac{\sigma_x^2}{n}\right)$; $\bar{y} \sim N\left(b; \frac{\sigma_y^2}{m}\right)$

$H_1: a-b>0$

$$\bar{x}-\bar{y} \sim N\left(a-b; \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

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$$\Rightarrow \Delta = \frac{(\bar{x}-\bar{y}) - \overset{0 \text{ при } H_0}{(a-b)}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0;1)$$

$$G = \{\Delta \geq u_{1-\alpha}\}$$

$$u_{0,95} \approx 1,645$$

3. $$\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}} = \sqrt{\frac{2}{3} + \frac{1}{2}} = \sqrt{\frac{7}{6}} \approx 1,08$$

$$\Delta = \frac{-1,597+2,6}{1,08} \approx 0,93 < u_{0,95} \Rightarrow$$

$\Rightarrow$ <u>нет оснований отвергать H_0</u>

4. Пусть $\delta = a-b$

При H_1:

$$\frac{(\bar{x}-\bar{y}) - \delta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0;1)$$

$$W(\delta) = P\left(\frac{\overset{\neq \delta}{\bar{x}-\bar{y}}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \geq u_{1-\alpha}\right) = P\left(\overset{\neq \delta}{\bar{x}-\bar{y}} \geq u_{1-\alpha}\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}\right)$$

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